

Machine Learning Interview Questions with Examples (Fresher)

Q: What is AI?

A: AI enables machines to perform tasks that normally require human intelligence. Example: Chatbots, recommendation systems.

Q: What is Machine Learning?

A: A subset of AI where machines learn patterns from data. Example: Predicting spam emails.

Q: Difference between AI, ML and Deep Learning?

A: AI is the broad field. ML is a subset of AI. Deep Learning is a subset of ML using neural networks.

Q: What is a Dataset?

A: A collection of data used for training/testing. Example: Iris flower dataset.

Q: What are Features?

A: Input variables used by a model. Example: Sepal length, petal width.

Q: What is a Label/Target?

A: The output the model predicts. Example: Setosa, Versicolor, Virginica.

Q: What is Training?

A: The process of teaching a model from data.

Q: What is Testing?

A: Evaluating a trained model on unseen data.

Q: What is Train-Test Split?

A: Dividing data into training and testing parts, e.g., 80% train and 20% test.

Q: What is Overfitting?

A: When a model memorizes training data and performs poorly on new data.

Q: What is Underfitting?

A: When a model fails to learn enough patterns from data.

Q: What is Accuracy?

A: Percentage of correct predictions made by the model.

Q: What is Classification?

A: Predicting categories. Example: Spam or Not Spam.

Q: What is Regression?

A: Predicting numerical values. Example: House prices.

Q: What is Random Forest?

A: A machine learning algorithm using multiple decision trees.

Q: What is a Decision Tree?

A: A tree-like model that makes decisions based on conditions.

Q: What is Logistic Regression?

A: A classification algorithm despite its name.

Q: What is Pickle?

A: A Python library used to save trained models.

Q: What is model.pkl?

A: A saved trained machine learning model.

Q: What is Pandas?

A: A Python library for data manipulation and analysis.

Q: What is NumPy?

A: A Python library for numerical computations and arrays.

Q: Why use Docker in ML?

A: To package code, model, Python, and dependencies together.

Q: What is a Docker Image?

A: A portable package containing the application and dependencies.

Q: What is a Docker Container?

A: A running instance of a Docker image.

Q: Explain your Iris Project.

A: I trained a Random Forest model on the Iris dataset, saved it with Pickle, and containerized it using Docker.

Q: Explain Spam Detection Project.

A: Text messages were converted into numerical vectors and classified as spam or not spam.

Q: How do you evaluate a model?

A: Using accuracy, precision, recall, F1-score, and confusion matrix.

Q: What is Precision?

A: Out of predicted positives, how many were actually positive.

Q: What is Recall?

A: Out of actual positives, how many were correctly identified.

Q: What is Confusion Matrix?

A: A table showing correct and incorrect predictions.